

[8] Connectors

DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

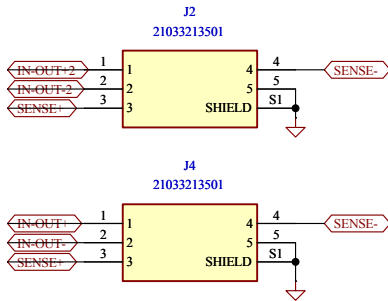
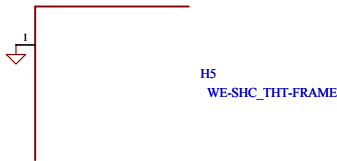
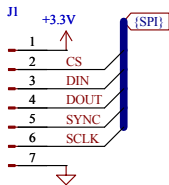
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,
(+3.3V and GND are connected to the PWR_FLAG and GND pins).



Host and load-cell connectors



Comments: References: JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A3	Revision: + (Unreleased)
			Sheet: 8	of 10